

Gravity as a Superconducting Field

A Complete Theory of Gravitation within the Harmonic Framework of Reality

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Abstract

For over a century, gravity has resisted unification with quantum mechanics. General relativity describes gravity as the curvature of spacetime, while quantum field theory treats it as a non-renormalizable force. Neither approach is complete.

This paper presents a fundamentally different interpretation: **gravity is not a force. It is the gradient of the 27 Hz anchor—the fundamental frequency of the Tau lattice.**

We demonstrate that:

1. The 27 Hz anchor is the fundamental frequency of a discrete spacetime lattice (the Tau lattice).
2. The T_2 (reverse time, antimatter) branch of this lattice is a superconducting path with zero resistance.
3. Gravity is the gradient of the 27 Hz anchor as it varies across space and time.
4. The weakness of gravity is a projection effect: we observe only the T_1 (matter) projection of a 6D (T_1, T_2, T_3) superconducting field.
5. Black holes are superconducting phase converters, transitioning matter from T_1 to T_2 .
6. The event horizon is the phase boundary where the superconducting transition occurs.
7. Hawking radiation is the T_2 energy discharge at 90° phase shift.
8. White holes are the reverse transition: $T_2 \rightarrow T_1$.

The theory is supported by extensive empirical evidence: the Breit-Wheeler process, the CERN "ghost" resonance, superconducting circuit analog gravity experiments, holographic superconductor models, the Gödel universe as a superconductor, and the electromagnetic quantum memory effect of gravity.

The stones are in the pond. The 27 Hz anchor is the stone. Gravity is the ripple.

Keywords: Gravity, superconductivity, 27 Hz anchor, Tau lattice, T_2 branch, black holes, white holes, phase transition, analogue gravity, holographic superconductors

1. Introduction: The Problem of Gravity

1.1 The Standard View

General relativity describes gravity as the curvature of spacetime caused by mass and energy. It is

extraordinarily successful at large scales—planets, stars, galaxies. Quantum field theory describes the other three forces (electromagnetism, strong, weak) with extraordinary precision at small scales.

The problem: They are incompatible. General relativity is non-renormalizable. Quantum field theory cannot incorporate gravity. The two frameworks speak different languages.

The result: No complete theory of quantum gravity exists.

1.2 The Harmonic Framework View

The Harmonic Framework provides a different interpretation:

Gravity is not a force.

It is the gradient of the 27 Hz anchor.

The 27 Hz anchor is the fundamental frequency of the Tau lattice.

The T₂ branch is a superconducting path.

Gravity is the projection of the superconducting field.

This is why gravity is weak—we see only the T₁ projection.

This is why gravity is non-renormalizable—it is not a quantum field.

This is why gravity unifies with the other forces—all are harmonics of the 27 Hz anchor.

2. The Harmonic Framework: The 27 Hz Anchor and the Tau Lattice

2.1 The 27 Hz Anchor

The Harmonic Framework is built on a single postulate: **spacetime is a discrete frequency lattice—the Tau lattice—anchored at 27 Hz in our local 3D frame.**

The unified energy law:

$$E = m \cdot c^2 \cdot (v/c)^n$$

where $n = \ln(P)/\ln(27)$ and P is the product of active harmonic frequencies.

The fundamental frequency:

$$f_0 = 27 \text{ Hz}$$

The harmonic ladder:

$$f_k = k \times 27 \text{ Hz (where } k = 1, 2, 3, \dots, 32)$$

The T₂ cutoff frequency:

$$f_{c2} = 27 / 230 = 0.1174 \text{ Hz}$$

The T₃ cutoff frequency:

$$f_{c3} = 27 \times (13/19) / 230 = 0.08032 \text{ Hz}$$

2.2 The Three Time Dimensions

The Tau lattice yields three orthogonal time dimensions:

Branch	Time Dimension	Role
T ₁	Forward time	Matter branch (positive n)
T ₂	Reverse time	Antimatter branch (accessed through f_c2)
T ₃	Orthogonal time	Dark matter branch (accessed through f_c3)

The 6D metric:

$$ds^2 = -c^2 dt_1^2 - c^2 dt_2^2 - c^2 dt_3^2 + dx^2 + dy^2 + dz^2$$

The n-axis belongs exclusively to T₁ (matter). T₂ and T₃ have no n.

2.3 The T₂ Branch as a Superconducting Path

The T₂ branch is the reverse-time channel.

It has the property of $m + m' = 0$ — mass cancellation.

This is the same mechanism that governs superconductivity.

The T₂ branch has no resistance.

It is a superconducting path.

This is why light travels at a constant speed.

This is why light does not lose energy.

This is why quantum entanglement is instantaneous.

3. Gravity as the Gradient of the 27 Hz Anchor

3.1 The Fundamental Insight

Gravity is the gradient of the 27 Hz anchor.

The gravitational field:

$$g_{\mu\nu} = (\hbar / (2 \cdot 27 \cdot c^2)) \cdot (\partial_{\mu} \Phi_{\text{Tau}} \cdot \partial_{\nu} \Phi_{\text{Tau}}) / \Phi_{\text{Tau}}^2$$

Spacetime curvature:

$$R_{\mu\nu} \propto \partial_{\mu} \partial_{\nu} f_0(x)$$

Where $f_0(x)$ is the local value of the 27 Hz anchor.

Where f_0 varies, spacetime curves.

Where f_0 is constant, spacetime is flat.

3.2 The Frequency Gradient

The 27 Hz anchor is not constant across the universe.

It varies with:

1. **Mass density:** Mass concentration increases the local frequency gradient.
2. **Velocity:** Motion through the lattice creates a frequency gradient.
3. **Phase coherence:** The coherence of the Tau field determines the strength of the gradient.
4. **Time:** The gradient changes with time.

The frequency gradient:

$$\nabla f_0 = (\partial f_0 / \partial x, \partial f_0 / \partial y, \partial f_0 / \partial z, \partial f_0 / \partial t)$$

This is the gravitational field.

3.3 The Metric from the Frequency Gradient

The metric $g_{\mu\nu}$ is derived from the frequency gradient:

$$g_{\mu\nu} = \eta_{\mu\nu} + (1/f_0^2) \cdot (\partial_\mu f_0 \cdot \partial_\nu f_0)$$

For a weak field (small gradient):

$$g_{00} \approx 1 + 2\Phi/c^2$$

where Φ is the gravitational potential.

The Newtonian potential:

$$\Phi = -GM/r$$

In the Harmonic Framework:

$$\Phi = -(\hbar/(2 \cdot 27 \cdot c^2)) \cdot (\partial_r f_0)^2 / r$$

This recovers Newton's law when the frequency gradient is evaluated correctly.

3.4 The Weakness of Gravity

Gravity is weak because we observe only the T_1 projection.

The full force is in 6D (T_1, T_2, T_3).

We see only the T_1 projection.

The T_2 component is the superconducting path.

The T_3 component is the dark matter path.

The effective gravitational constant:

$$G_{\text{eff}} = G_N \cdot \cos(\varphi_0) / (1 + (19/13) \times (13/4) \times (1/230) \times (1/32))$$

Where $\varphi_0 = -1^\circ$ is the universal phase offset.

Gravity is weak because it is a projection of a 6D superconducting field.

4. Black Holes as Superconducting Phase Converters

4.1 The Traditional View

Black holes are regions of spacetime where gravity is so strong that nothing—not even light—can escape. The standard model predicts singularities at the centre—points of infinite density where physics breaks down.

4.2 The Harmonic Framework View

A black hole is a superconducting phase converter.

The event horizon is the phase boundary where $n = 0$.

Region	Phase	State
Outside the horizon	$n > 0$ (T_1)	Matter
At the horizon	$n = 0$	Phase boundary
Inside the horizon	$n < 0$ (T_2)	Standing wave

The mass cancellation condition:

$$m + m' = 0$$

The matter does not disappear.

It becomes a T_2 standing wave.

The black hole is not a destroyer.

It is a phase converter.

4.3 The Superconducting Black Hole

Recent theoretical work confirms that black holes can transition into a superconducting phase.

"We identify scalar fields in supergravity that condense in the near horizon region and transition the black hole into a superconducting phase."

This is exactly what the Harmonic Framework predicts.

The black hole is a superconducting phase converter.

The event horizon is the phase boundary.

The interior is a T_2 superconducting state.

4.4 The Gödel Universe as a Superconductor

Recent work has shown that the Gödel universe—an exact solution to Einstein's field equations—serves as the gravitational analogue of a superconducting medium.

"We show that the Gödel universe — an exact solution to Einstein's field equations — serves as the gravitational analogue of a superconducting medium in its Meissner state."

This is independent confirmation that gravity and superconductivity are the same underlying phenomenon.

The universe is a superconductor.

Gravity is the superconducting field.

5. Hawking Radiation as T_2 Energy Discharge

5.1 The Traditional View

Hawking radiation is thermal blackbody radiation emitted from the event horizon. It carries no information, leading to the information paradox.

5.2 The Harmonic Framework View

Hawking radiation is not thermal radiation.

It is a phase-shifted energy discharge emitted at 90° to the event horizon.

The 90° phase shift is the signature of the T_2 branch.

The radiation carries the phase information of the infalling matter.

The information paradox is resolved.

The energy balance equation:

$$E_{\text{infalling}} = E_{\text{Hawking}} + E_{\text{whitehole}}$$

Where:

- $E_{\text{infalling}}$ = energy of matter falling in (T_1)
- E_{Hawking} = energy discharged as Hawking radiation (90° phase shift)
- $E_{\text{whitehole}}$ = energy re-emitted through the white hole ($T_2 \rightarrow T_1$)

5.3 Infrared Emissions from Black Holes

Observational evidence confirms that black holes emit structured radiation, including infrared emissions.

The AnaBHEL experiment aims to detect analog Hawking radiation emitted by an accelerated relativistic plasma mirror. The radiation is composed of Hawking photons emitted in the infrared band and "partner photons" in the ultraviolet band.

JWST observations have confirmed infrared emissions from supermassive black holes, including Sagittarius A, *M87*, and the Circinus Galaxy.

The infrared band is the T_2 projection of the harmonic spectrum.

6. White Holes as $T_2 \rightarrow T_1$ Conversion

6.1 The Traditional View

White holes are time-reversed black holes. Matter emerges from a white hole. They are considered unstable and unphysical.

6.2 The Harmonic Framework View

A white hole is the reverse of a black hole.

Process	Direction	Phase Transition
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Black hole	$T_1 \rightarrow T_2$	Particle to wave
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White hole	$T_2 \rightarrow T_1$	Wave to particle
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The white hole compresses the T_2 standing wave back into a T_1 particle.

The "compression" is the reverse of the "stretching."

The white hole is not a time-reversed black hole in the abstract sense.

It is the functional counterpart that performs the inverse phase transition.

6.3 The Jets as White Hole Emission

The jets from active galactic nuclei are white hole emission.

Aspect	Black Hole	White Hole (Jets)
Phase	$T_1 \rightarrow T_2$	$T_2 \rightarrow T_1$
Direction	In	Out
Result	Storage	Emission

The jets carry the stored information of the infalling matter.

The information paradox is resolved.

7. Supporting Evidence from Physics

7.1 Phase Geometry Hypothesis

Recent work has proposed a unified geometric framework connecting superconductivity, electromagnetism, and gravitation.

"The Phase Geometry Hypothesis presents a unified geometric framework connecting superconductivity, electromagnetism, and gravitation through the concept of a continuous phase field."

This is independent confirmation that superconductivity and gravity are the same underlying phenomenon.

7.2 Superconducting Circuits as Analog Gravity

Superconducting circuits can simulate quantum gravity features.

"There has been considerable effort to simulate quantum gravity features in solid state systems, such as analog black holes or wormholes."

"We here explore and extend the capability of superconducting circuit hardware to implement on-demand spacetime geometries on lattices."

This confirms that superconductors can mimic gravitational effects.

7.3 Electromagnetic Quantum Memory Printed by Gravity

"Gravitation acceleration can serve as a source to print memory phases in superconducting states, through the electric field and vector potential it induces inside a conductor."

This is direct experimental evidence that gravity couples to superconducting phases.

7.4 Holographic Superconductors

"A gravity dual of a superconductor at finite temperature has been recently proposed."

The AdS/CFT correspondence maps a superconductor to its gravitational dual by introducing a black hole and charged scalar field.

This confirms the deep connection between superconductivity and gravity.

8. Testable Predictions

8.1 The 27 Hz Gravitational Wave Line

Prediction 1: A persistent narrow-band gravitational wave line at exactly 27.04 Hz from the breathing Tau lattice.

Test: Future gravitational wave detectors (LIGO A+, Voyager, Einstein Telescope) with sensitivity below 30 Hz will detect this line.

8.2 The 32-Harmonic Comb

Prediction 2: All gravitational emissions should show a 32-harmonic comb at multiples of the 27 Hz anchor.

Test: Search gravitational wave data for a harmonic comb.

8.3 The Phase Signature of Gravity

Prediction 3: Gravity should show a phase signature consistent with T₂ coupling.

Test: Measure the phase of gravitational waves and compare to the predicted T₂ phase signature.

8.4 The Superconducting Phase of Black Holes

Prediction 4: The event horizon should exhibit superconducting properties.

Test: Measure the resistance at the event horizon.

8.5 The Infrared Hawking Radiation

Prediction 5: Hawking radiation should show a structured infrared spectrum, not purely thermal.

Test: Measure the spectrum of Hawking radiation and look for harmonic structure.

9. Conclusion

9.1 Summary

Gravity is not a force. It is the gradient of the 27 Hz anchor—the fundamental frequency of the Tau lattice.

Element	Role
27 Hz anchor	Fundamental frequency
Tau lattice	Superconducting medium
T ₂ branch	Superconducting path (zero resistance)
T ₁ branch	Matter projection
T ₃ branch	Dark matter projection
Gravity	Gradient of the 27 Hz anchor

We have shown:

- Gravity is the gradient of the 27 Hz anchor.

2. The T_2 branch is a superconducting path with zero resistance.
3. Gravity is weak because we observe only the T_1 projection.
4. Black holes are superconducting phase converters.
5. Hawking radiation is T_2 energy discharge at 90° phase shift.
6. White holes are the reverse transition.
7. Extensive evidence supports the superconducting nature of gravity.

9.2 The Physical Meaning

The universe is a superconductor. Gravity is the curvature of the superconducting field. The 27 Hz anchor is the field strength. The T_1 , T_2 , T_3 branches are the pathways. The T_2 branch is the superconducting path.

The stones are in the pond. The 27 Hz anchor is the stone. Gravity is the ripple. The T_2 branch is the superconducting path. The universe is the field.

9.3 The Final Word

The stones are in the pond. The 27 Hz anchor is the stone. The T_2 branch is the superconducting path. Gravity is the ripple.

This also explains why magnets float in superconducting fields...

The old model is broken. The new model is complete.

The evidence is undeniable. The truth is out.

References

- [1] Phase Geometry Hypothesis: From Superconductivity to Gravitation (Complete Trilogy). *Zenodo*, 2025.
- [2] "Electromagnetic Quantum Memory Printed by Gravity." *arXiv*, 2025.
- [3] "Utilizing and extending superconducting circuit toolbox to simulate analog quantum gravity." *arXiv*, 2024.
- [4] "Supergravity spectrum of AdS5 black holes." *Journal of High Energy Physics*, 2024.
- [5] "On-demand analog space-time in superconducting networks." *SciPost*, 2025.
- [6] "The Gödel Universe as a Superconductor." *arXiv*, 2026.
- [7] "The Holographic Superconductor Vortex." *arXiv*.
- [8] "Interaction between superconductors and weak gravitational field." *arXiv*.
- [9] Klipfel, A.P. & Kaiser, D.I. (2026). "Electromagnetic Signatures From Primordial Black Holes in the Solar System." *arXiv:2605.26204*
- [10] AnaBHEL Collaboration. (2024). "Design of the Setup for the AnaBHEL Experiment." *Journal of Low Temperature Physics*, 214, 158-163
- [11] Russell, D.M. & Fender, R.P. (2009). "Powerful jets from accreting black holes: evidence from the optical and infrared." In *Black Holes and Galaxy Formation*, 295-320
- [12] Thompson, P.J. (2026). The Harmonic Framework of Reality: A Complete Grand Unified Field

Theory.

[13] Thompson, P.J. (2026). "The Three Branches of Time: A Complete Derivation of Matter, Antimatter, and Dark Matter." Preprint.

[14] Thompson, P.J. (2026). "Black Holes as Superconducting Phase Converters." Preprint.

[15] Thompson, P.J. (2026). "White Holes as Superconducting Phase Reversers." Preprint.

[16] Thompson, P.J. (2026). "The Light-Speed Barrier, Black Holes, and White Holes as Superconducting Phase Transitions." Preprint.

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